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Three-dimensional acoustic metamaterial Luneburg lenses for broadband and wide-angle underwater ultrasound imaging

Jung-Woo Kim [a,1], Gunn Hwang [b,*,1], Seong-Jin Lee [c], Sang-Hoon Kim [c], Semyung Wang [a,*]

[a] School of Mechanical Engineering, Gwangju Institute of Science and Technology, 123 Cheomdangwagi-ro, Buk-gu, Gwangju 61005, Republic of Korea
[b] Brain Links Creative Research Section, Future and Basic Technology Research Division, Electronics and Telecommunications Research Institute, 218 Gajeong-ro, Yuseong-gu, Daejeon 34129, Republic of Korea
[c] Division of Marine Engineering, Mokpo National Maritime University, 91 Haeyangdaehak-ro, Mokpo-si, Jeollanam-do 58628, Republic of Korea

A R T I C L E I N F O

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A B S T R A C T

This paper introduces spherical and ellipsoidal metamaterial lenses designed based on the Luneburg lens and transformation optics for underwater ultrasound imaging, which enhances signals over a broadband frequency range and detects objects at a wide-angle. The meta-atoms were analyzed with dispersion curves, equi-frequency contours, and effective acoustic parameters to design the lenses and predict the focusing performance. The three-dimensional lenses made of stainless steel were demonstrated numerically and experimentally in an underwater environment to determine the focal length and full width at half maximum. The spherical lens is omnidirectional. The focal length and the full width at half maximum of the spherical lens are 0λ and 0.6λ at 80 kHz, respectively. The ellipsoidal lens is thinner on the x-axis than that of the spherical lens. This lens has broadband properties by achieving the focal length and the full width at half maximum of 1.5λ to 0λ and 0.95λ to 0.59λ at frequencies between 60 kHz and 160 kHz, respectively. These lenses can potentially be applied to sound navigation ranging or medical transducers, providing the opportunities to view the underwater world with improved resolution at broadband and wide-angle.

1. Introduction

An aberration-free spherical Luneburg lens [1] is a gradient index (GRIN) lens that concentrates propagating waves to the opposite surface of the lens. The lens was modified to generate an inner focus and an outer focus [2], and it was numerically generalized [3]. Although the lens is not perfect [4] because of wavelength limitations [5], it exhibits advantages over conventional lenses because its focus does not suffer from aberrations at all incident angles. The applicability of the lenses in areas such as radar detection and antennas that require wide-angle scanning and high-gain radiation performance [6–12] have been widely researched owing to the excellent focusing, beam steering, and collimation capabilities.

With the advent of metamaterials, extensions into the acoustic realm, and applications of transformation optics (TO) [13–22], the

* Corresponding authors.
E-mail addresses: hwangun@etri.re.kr (G. Hwang), smwang@gist.ac.kr (S. Wang).
[1] Equal contribution as co-first authors.

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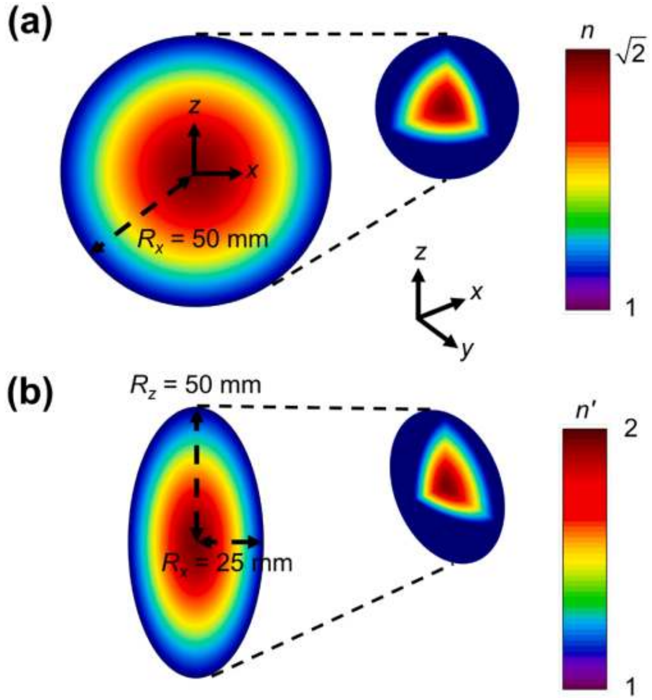

Fig. 1. Refractive index distribution of the (a) spherical and (b) ellipsoidal lens. The radius of the spherical lens is 50 mm and that of the ellipsoidal lens based on TO is 25 mm in the *x*-axis and 50 mm in the *y*- and z-axes. The lenses satisfy the refractive indices 1 to $\sqrt{2}$ and 1 to 2, respectively. The color bars indicate the refractive index ranges and the insets illustrate the 3D distributions.

structure and performance have become versatile and improved since the lenses were first proposed. Further, research on GRIN devices with phononic crystals (PC) has inspired interest in focusing lenses [23–29], and two- (2D) and three-dimensional (3D) acoustic metamaterial Luneburg lenses have been developed in air and water environments, respectively [30–38]. Moreover, structural GRIN lenses have been studied [39–41]. Similar to Luneburg lenses, GRIN PC flat lenses focus acoustic waves by changing the refractive index [24–26]. Unlike superlenses [4], which can achieve subwavelength imaging because of negative refraction, the Luneburg lenses exhibit positive material properties. The inclusion structures, which determine the degree of anisotropy and effective parameters, are suitable for broadband applications because of the nonresonant properties [27]. The main advantage of acoustic lenses over optical and electromagnetic lenses is that the lenses can be used in a water environment [26,28,33].

Ultrasonic transducers have advantages of low radiation exposure, portability, low cost compared to other imaging systems, and real-time inspection [43]. However, there is a risk of misdiagnosis because the ultrasound imaging performance varies based on the direction of the transducer. Further, it cannot detect objects smaller than the wavelength of ultrasonic waves, limiting the resolution. It is rare for GRIN metamaterial lenses to be implemented in water [32,33] or ultrasonic domains [42] as compared to the acoustic domain in air. Thus, the underwater acoustic lenses that can obtain ultrasound imaging in a wide field of view (FOV) should be studied to address these problems [35,43–45].

This study introduces spherical and ellipsoidal GRIN lenses developed using periodic composite materials, i.e., metamaterials, to achieve underwater ultrasound imaging. The metallic 3D Luneburg lenses with 10 concentric layers were designed. The refractive indices were achieved by adjusting the mass density of the scatterers, that is, meta-atoms or inclusions. The ellipsoidal lens that was designed based on TO was realized; given the lower volume, it is economical in terms of space. The scatterers were characterized by dispersion curves, effective refractive indices, and equi-frequency contours (EFCs) or equi-frequency surfaces (EFSs calculated using PC band structure calculations. Further, the scattering parameter (S-parameter) retrieval method was introduced to discuss the effective acoustic parameters of scatterers in detail. The spherical lens showed better beam steering performance than the ellipsoidal lens numerically and experimentally, based on comparisons of the focal length and the full width at half maximum (FWHM). The ellipsoidal lens (at frequencies within 60 kHz to 160 kHz) was operated at higher and broader frequency range than that of the spherical lens (at frequencies within 30 kHz to 100 kHz).

2. Refractive indices and TO

An acoustic spherical Luneburg lens focuses acoustic waves or transforms point sources into plane waves without aberration in all directions. However, it is not easy to apply this lens directly to devices owing to the large volume. Therefore, a spherical Luneburg lens, which is attributed a refractive index of

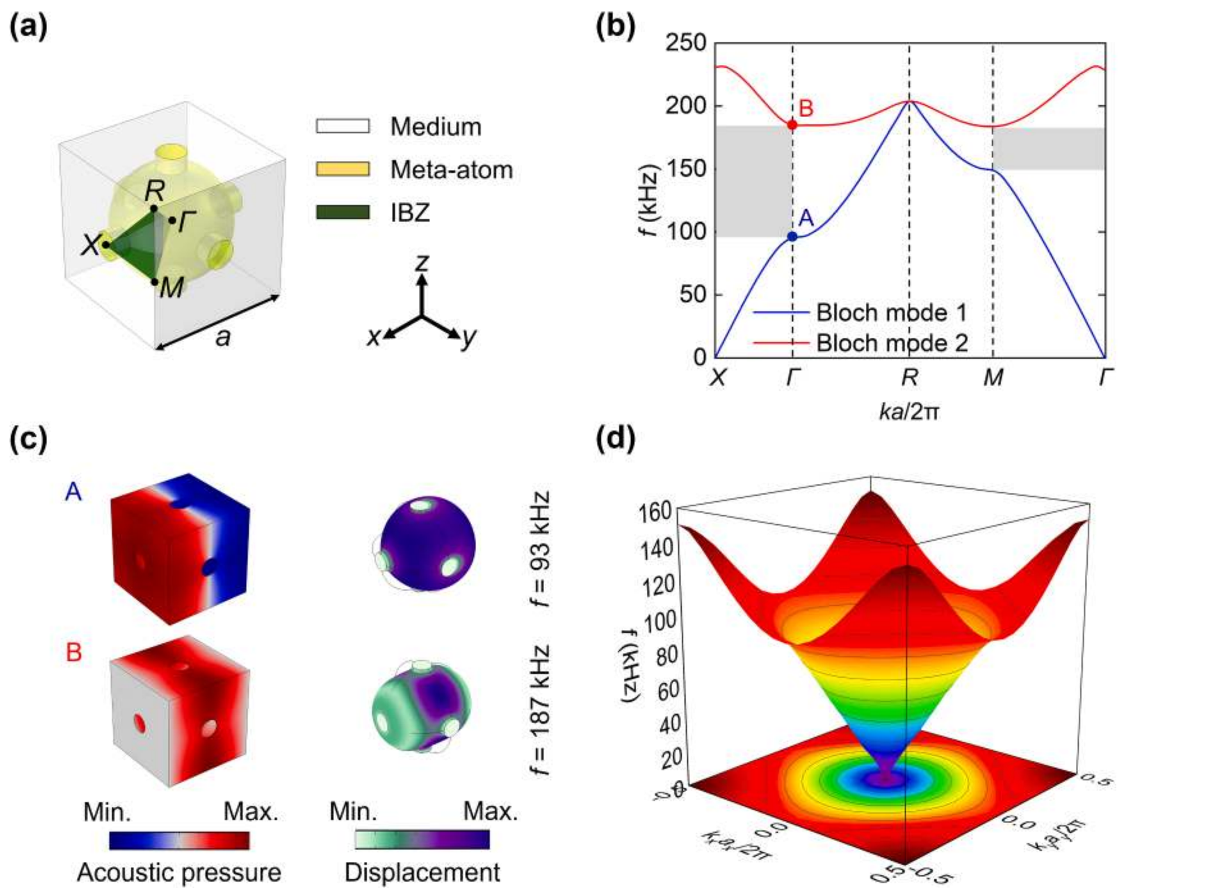

Fig. 2. (a) PC band structure of the spherical meta-atom combined with the cylindrical supporters in the x-, y-, and z-axes directions. (b) Bloch modes and PBG. (c) Numerical simulation results for the acoustic pressure and displacement of the PC band structure. (d) EFC and its projection.

$$n(r) = \sqrt{2 - \left(\frac{r}{R}\right)^2}, \tag{1}$$

where $n(r)$ represents the refractive index of the radial distance, $r = \sqrt{x^2 + y^2 + z^2}$, and R denotes the radius of the lens [1], can be transformed into an ellipsoidal lens based on TO without affecting its properties [19,21,22]; the refractive indices are calculated, as shown in Fig. 1. The spherical lens has a radius $R = 50$ mm and that of the ellipsoidal lens is halved along the x-axis direction, that is, $R_x = 25$ mm.

The transformation formula is defined to reduce and compress the radius of the x-axis direction as $x' = (\delta/R)x$, $y' = y$, and $z' = z$, where δ represents the compression factor and is an any real integer [16]. A spatially varying mass density and refractive index are expressed as

$$\rho(r) = 2 - r^2/R^2 = n^2(r), \tag{2}$$

and the lenses are supposed to be incompressible (bulk moduli B and B' are assumed to be approximately equal to 1 in this case). The mass density of the ellipsoidal lens can be expressed as a Jacobian matrix J combined as

$$\overline{\overline{\rho}}' = \frac{J\overline{\overline{\rho}}J^T}{\det(J)}, \tag{3}$$

where $\overline{\overline{\rho}}$ and $\overline{\overline{\rho}}'$ represent the mass density tensors of the spherical and ellipsoidal lenses; J^T and $\det(J)$ represent the transpose and determinant of J, respectively [16,21]. When the effective mass density is less than 1, the meta-atoms resonate, which results in a narrower frequency bandwidth and increased energy loss. Thus, ρ'_{yy} and ρ'_{zz} are selected as 1 without affecting the imaging performances by focusing and avoiding the high and low refractive index regions of ρ'-map; however, there are exceptions for some particular incident angles, such as $\pm 90^\circ$[14–16]. Therefore, the transformed mass density [16,17,20–22] can be obtained as

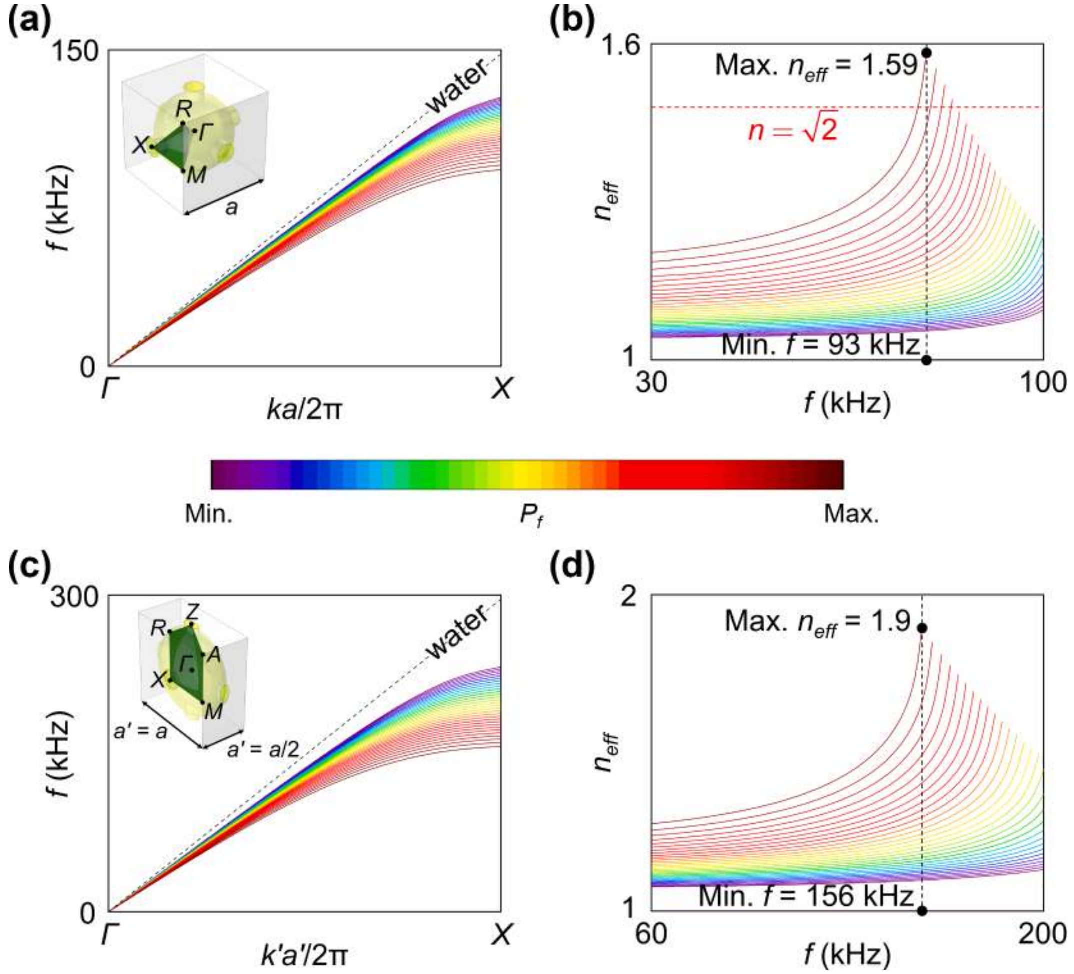

Fig. 3. Dispersion curves along the 1st acoustic band of the (a) spherical and (c) ellipsoidal meta-atoms according to packing factor P_f. Effective refractive indices of the (b) spherical and (d) ellipsoidal meta-atoms derived by the dispersion relation.

$$\rho'_{xx} = \delta\left[2 - \frac{(\delta x')^2 + y^{2'} + z^{2'}}{R^2}\right]. \tag{4}$$

Here, as $\delta = 2$, the radius of the ellipsoidal lens in the x-axis direction is halved for that of the spherical lens. The refractive indices of the spherical and ellipsoidal lenses are derived from the formulas range from 1 to $\sqrt{2}$ and 1 to 2, as shown in Figs. 1(a) and (b), respectively. This method and these assumptions lead to the realization of broadband metamaterial lenses as the meta-atoms are nondispersive. Further, this makes the thickness of the ellipsoidal lens thinner than that of the spherical lens, resulting in more efficient lenses and providing insights for acoustic devices in applications such as medical ultrasonography, nondestructive evaluation, and sound navigation ranging.

3. Properties of meta-atoms

3.1. Dispersion relation and EFCs

It is necessary to calculate the effective acoustic parameters of meta-atoms to be placed at each layer because the lenses are discretized to satisfy the refractive indices (or mass densities). The PC band structure calculations [46–48] in Fig. 2 and S-parameter retrieval method [49–53] in Fig. 5 were introduced to analyze and predict the imaging performance. Multiple scattering theory (MST) requires loosely-packed inclusion condition. The theory was excluded in this study because it is not suitable for calculating the refractive indices especially around the center of the lenses where the inclusion is very tightly-packed [54,55].

The commercial finite element method software, COMSOL Multiphysics V5.3, was used to simulate the multiphysics coupled with acoustics and structural mechanics. PC band structure calculations were performed by eigenfrequency analysis. Floquet-Bloch boundary condition was applied to the PC band structure in Fig. 2(a). Irreducible Brillouin zone (IBZ) path of the PC band

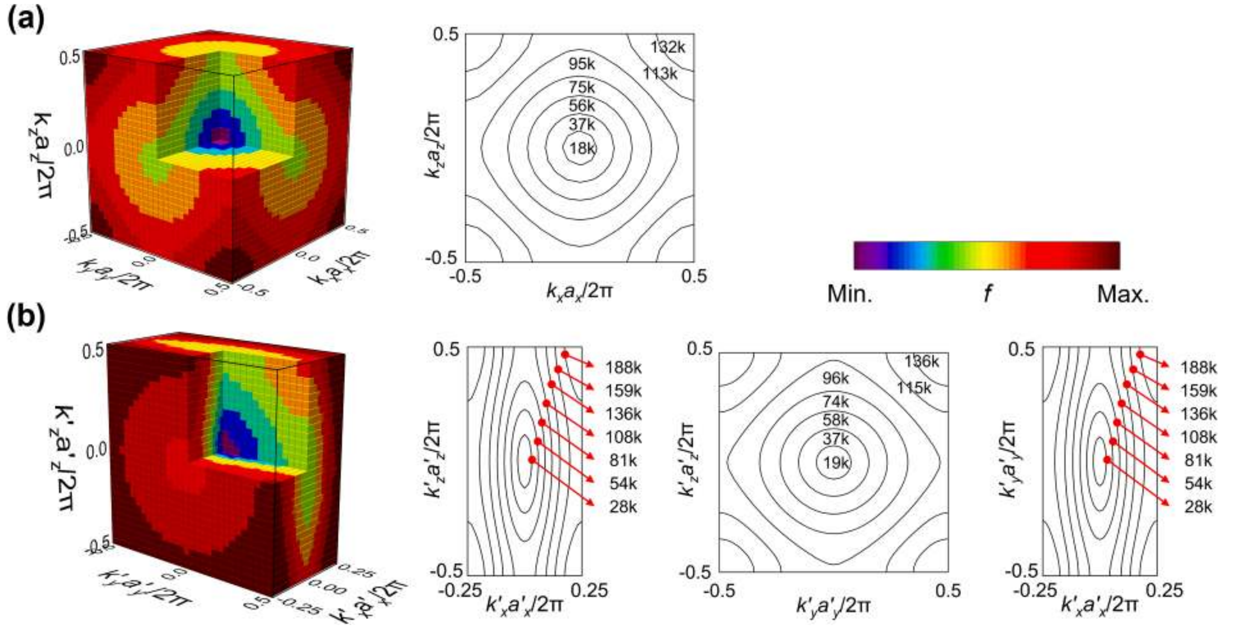

Fig. 4. EFCs (or EFSs) using the PC band structure calculations. EFSs and EFCs of the bulkiest (a) spherical and (b) ellipsoidal meta-atoms at the center of the lenses.

structure is as follows: X–Γ–R–M–Γ. White, yellow, and green represent medium, meta-atom, and IBZ, respectively. Phononic band gap (PBG), Bloch modes (1 and 2), and EFC (and its projection) were demonstrated, as shown in Figs. 2(b)–(d). The simulations are performed on the largest spherical meta-atom. The lattice constant a is 5 mm, and the diameter of the spherical meta-atom is 4.9 mm. The PBG along the X–Γ direction, which blocks wave propagation, was formed between in the frequency range of 93 kHz to 187 kHz, and A (blue point) and B (red point) are indicated at the ends of the dispersion curve. Bloch mode 2 corresponding to PBG was obtained in B. EFCs support these results because of the homogeneous frequency range, as shown in Fig. 2(d).

Stainless steel ($\rho = 8{,}000$ kg/m^3, $E = 200$ GPa, and $\nu = 0.27$) was selected as the material of the lenses for the numerical simulations and the experiments. The meta-atoms are combined with cylindrical supporters in the x-, y-, and z-axes directions (radius = 0.5 mm). Similarly, the ellipsoidal meta-atoms are joined by the cylindrical supporters whose radius is 0.25 mm in the x-axis and the elliptical supporters of which the cross-sectional axis lengths are 0.5 mm in the y- and z-axes. The 3D structures were directly fabricated with a selective laser sintering 3D printer because of these supporters.

In Fig. 3, the dispersion curves and effective refractive indices were calculated using the PC band structure calculations in Fig. 2. The lattice constant of the simple cubic lattice (CUB) comprising the spherical lens is 5 mm. The length in the x-axis direction of the tetragonal lattice (TET) is 2.5 mm, which is half of the lattice constant of the CUB. The insets of Figs. 3(a) and (c) show that the dispersion curves of the 1st acoustic band were obtained. The insets show PC band structures with meta-atoms (yellow) immersed in water (white); green parts represent the IBZ of the CUB and TET. The inclusions were calculated by sweeping the packing factor, which is defined as $P_f = N_{\text{atom}} V_{\text{atom}} / V_{\text{unit cell}}$ where N_{atom} is the number of the meta-atom, V_{atom} is the volume of the meta-atom, and $V_{\text{unit cell}}$ is the volume of the unit cell, i.e., it means the volume fraction of the crystal structure occupied by the inclusions. The packing factors of the meta-atoms are changed based on the radius of the Luneburg lens refractive index formula. Therefore, the minimum and maximum values of P_f of the spherical lens are 0.113 and 0.493, respectively. The black dashed lines represent the energy bands in water where there is no dispersion during wave propagation.

The dispersion relation in a homogeneous medium is expressed as

$$\omega(k) = \frac{c}{n_{eff}} \times k, \tag{5}$$

where $\omega(k)$ is the angular frequency dependent on the wavenumber k, c is the speed of sound in water, and n_{eff} is the effective refractive index. Based on this relation, n_{eff} was calculated, as shown in Figs. 3(b) and (d). The black points represent the maximum refractive index and the minimum frequency. The red dashed line represents the refractive index of the Luneburg lens. The higher the packing factor, the narrower the frequency band and the higher the effective refractive indices. The n_{eff} of the meta-atom at each layer were determined so that the lens was satisfied the effective refractive index. However, the ellipsoidal meta-atoms did not satisfy the refractive index of 2; therefore, it was designed to satisfy the maximum effective refractive index of 1.9. In the region where there is no line corresponding to the PBG frequency of each meta-atom, the refractive indices were excluded from the graph.

The frequency range is considered until the dispersion curve is nearly linear because the spherical lens is homogeneous and isotropic. It is necessary to closely examine the focus characteristics because the ellipsoidal lens is not homogeneous and isotropic. The

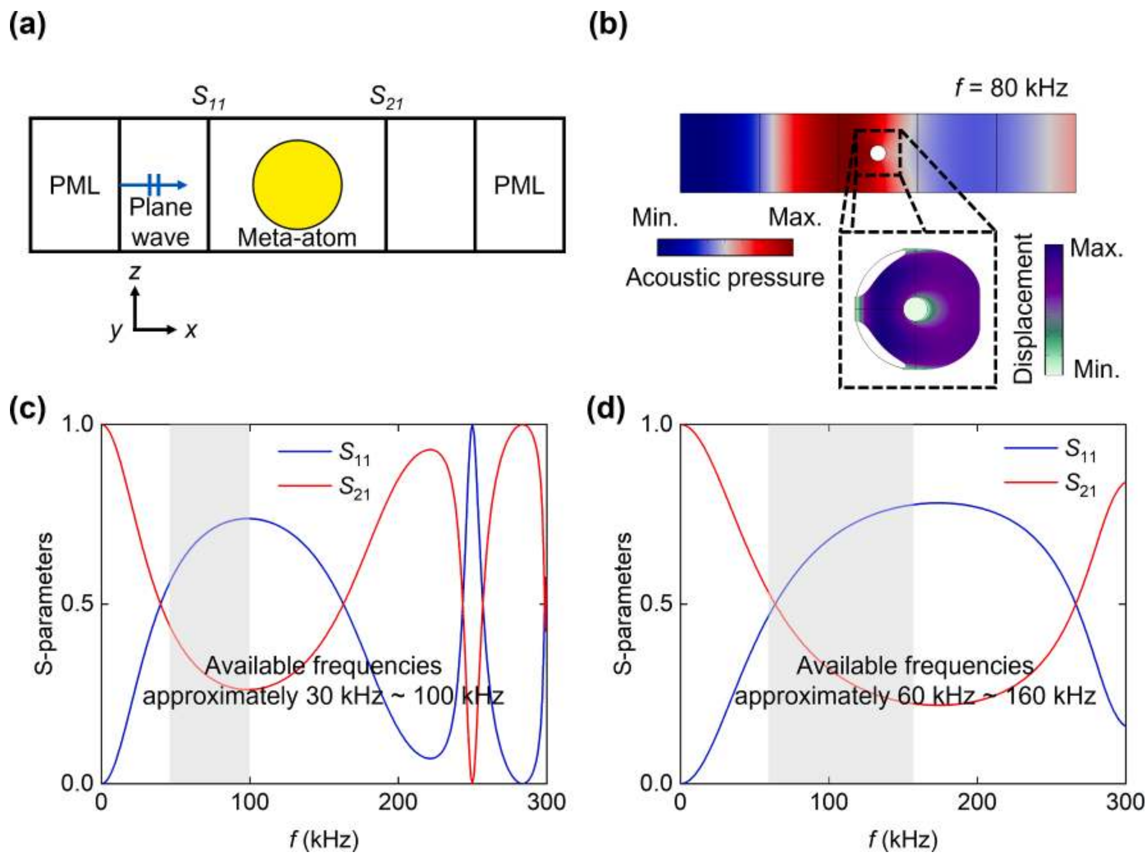

Fig. 5. (a) Schematic diagram of the numerical simulations for the S-parameter retrieval method. (b) Bloch mode of the spherical meta-atom by the method at a frequency of 80 kHz. Frequency-dependent S-parameters of the (c) spherical and (d) ellipsoidal meta-atoms.

EFCs and EFSs are analyzed to determine the behavior of the wave governed by the dispersion relations, as shown in Figs. 4(a) and (b). The EFCs are curves connecting the wavevectors of the propagating modes and demonstrate the focusing (or imaging) because the waves are determined by the shape of the curves. Therefore, the imaging performance is related to the slope and curvature of the EFCs in the wavevector $\boldsymbol{k}$ space for transmitting waves along a specific direction. Further, the contours are calculated using the PC band structure calculations, in which case the entire structures (CUB and TET) are scanned.

Group velocity is introduced to analyze the EFCs and defined as

$$\boldsymbol{v}_g \equiv \nabla_k \omega(\boldsymbol{k}), \tag{6}$$

which is perpendicular to the EFCs in a lossless medium and specified the direction of energy flow for the plane wave. The wave diffraction and propagation patterns based on the shapes of the EFCs are divided into zero, normal (positive), and anti (negative) diffractions [56]. Each shape is parallel, convex, and concave in the direction of the wave propagation, respectively. The EFCs of the spherical meta-atom are homogeneous up to a frequency of 95 kHz in Fig. 4(a), whereas the EFCs of the ellipsoidal one is not homogeneous from 108 kHz to 188 kHz in the k'_x vs. k'_z and k'_x vs. k'_y space, as shown in Fig. 4(b). However, the EFCs of the ellipsoidal meta-atom exhibit positive diffractions up to 159 kHz and anti-diffractions from 188 kHz. The minimum frequencies for the spherical and ellipsoidal meta-atoms with maximum refractive indices of 1.59 and 1.9, respectively, are 93 kHz and 156 kHz. The upper and lower limits of available frequencies for the spherical and ellipsoidal lenses are 100 kHz and 160 kHz, and 30 kHz and 60 kHz, respectively, because the wavelengths of these frequencies are smaller than the radius of the lenses for interacting with the acoustic waves.

3.2. Retrieved values

The effective acoustic parameters were calculated using the S-parameter retrieval method in the frequency domain. Fig. 5(a) shows the schematic diagram of the numerical simulations. Acoustic plane waves are normally incident on the scatterers from left to right, and perfectly matched layer (PML), which cancel out reflection waves, are set at both ends. The S-parameters are calculated at the boundary of the PC band structure. S_{11} and S_{21} refer to the reflection (R) and transmission coefficients (T), respectively. Bloch mode of the spherical meta-atom was calculated at a frequency of 80 kHz by the method and included as a representative example in Fig. 5(b).

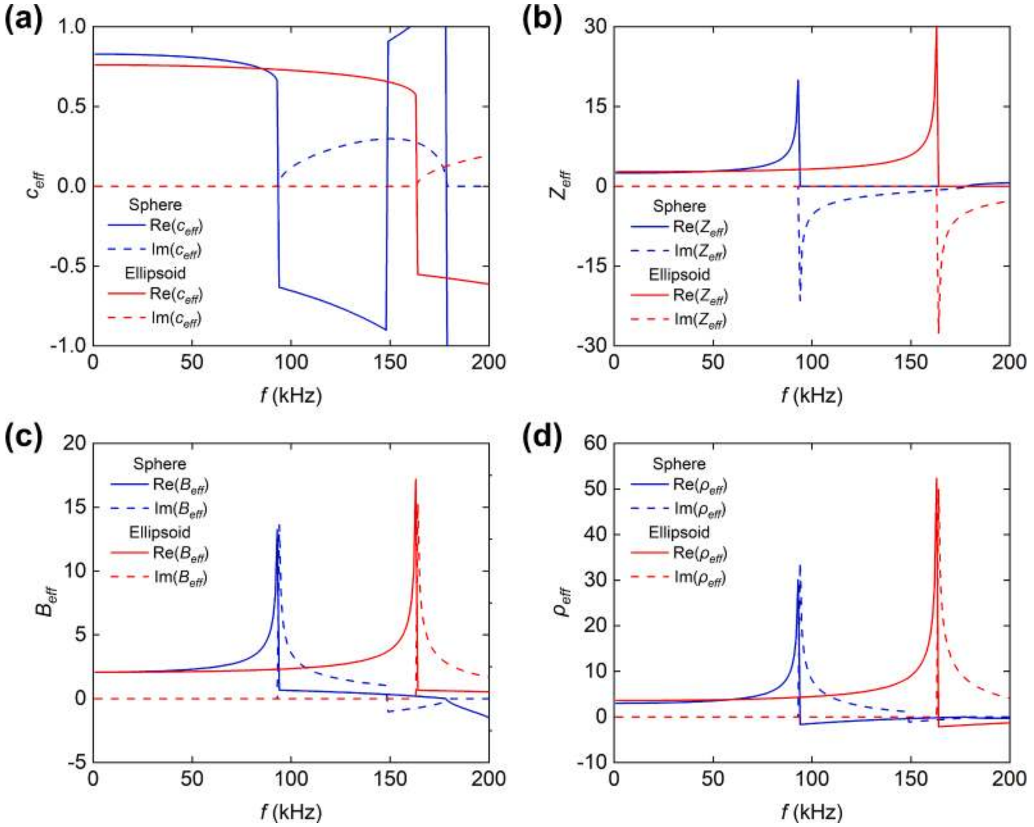

Fig. 6. Effective acoustic parameters using the S-parameter retrieval method. Real (solid line) and imaginary parts (dashed line) of (a) the effective speed of sound c_{eff}, (b) acoustic impedance Z_{eff}, (c) bulk modulus B_{eff}, and (d) mass density ρ_{eff} of the spherical (blue) and ellipsoidal meta-atoms (red) in the frequency range of 1 kHz to 200 kHz.

In Figs. 5(c) and (d), these graphs show the numerical simulation results for the spherical and ellipsoidal meta-atoms at frequencies from 1 kHz to 300 kHz. The available frequency ranges of the spherical and ellipsoidal meta-atoms are from 30 kHz to 100 kHz and 60 kHz to 160 kHz, respectively. The retrieved values are normalized. The reflection of the acoustic wave is greater than the transmission, which in turn is useful for verifying the frequency range for positive acoustic metamaterials.

Further, to verify the function of the lenses, the S-parameter retrieval method, which is extended from the effective parameters of electromagnetic materials [49–52] to acoustic metamaterials [52], was performed by normally incident plane waves on the PC band structure, and by calculating S_{11} and S_{21}, which are reflection and transmission coefficients, respectively in Fig. 5. The effective refractive index and acoustic impedance are expressed as

$$\begin{cases} n_{eff} = \pm\dfrac{\cos^{-1}(\{1-(S_{11}^2-S_{21}^2)\}/2S_{21})+2\pi m}{ka} \\ Z_{eff} = \pm\sqrt{\dfrac{(1+S_{11})^2-S_{21}^2}{(1-S_{11})^2-S_{21}^2}} \end{cases}, \tag{7}$$

where m represents the branch number of $\cos^{-1}$, k denotes the wavenumber, a represents the slab thickness (or lattice constant), and "±" restricts the imaginary part of the index Im(n_{eff}) to be negative [52]. When Re(Z_{eff}) or Im(n_{eff}) is close to zero, a computational error can cause incorrect combinations of the signs. Thus, it is redefined as

$$\begin{cases} n_{eff} = \dfrac{-i\log x+2\pi m}{kd} \\ Z_{eff} = \dfrac{r}{1-2S_{11}+S_{11}^2-S_{21}^2} \end{cases}, \tag{8}$$

where $x = (1-S_{11}^2+S_{21}^2+r)/2S_{21}, r = \mp\sqrt{(S_{11}^2-S_{21}^2-1)^2-4S_{21}^2}$, and the effective bulk moduli B_{eff} and the effective mass densities ρ_{eff} are defined as

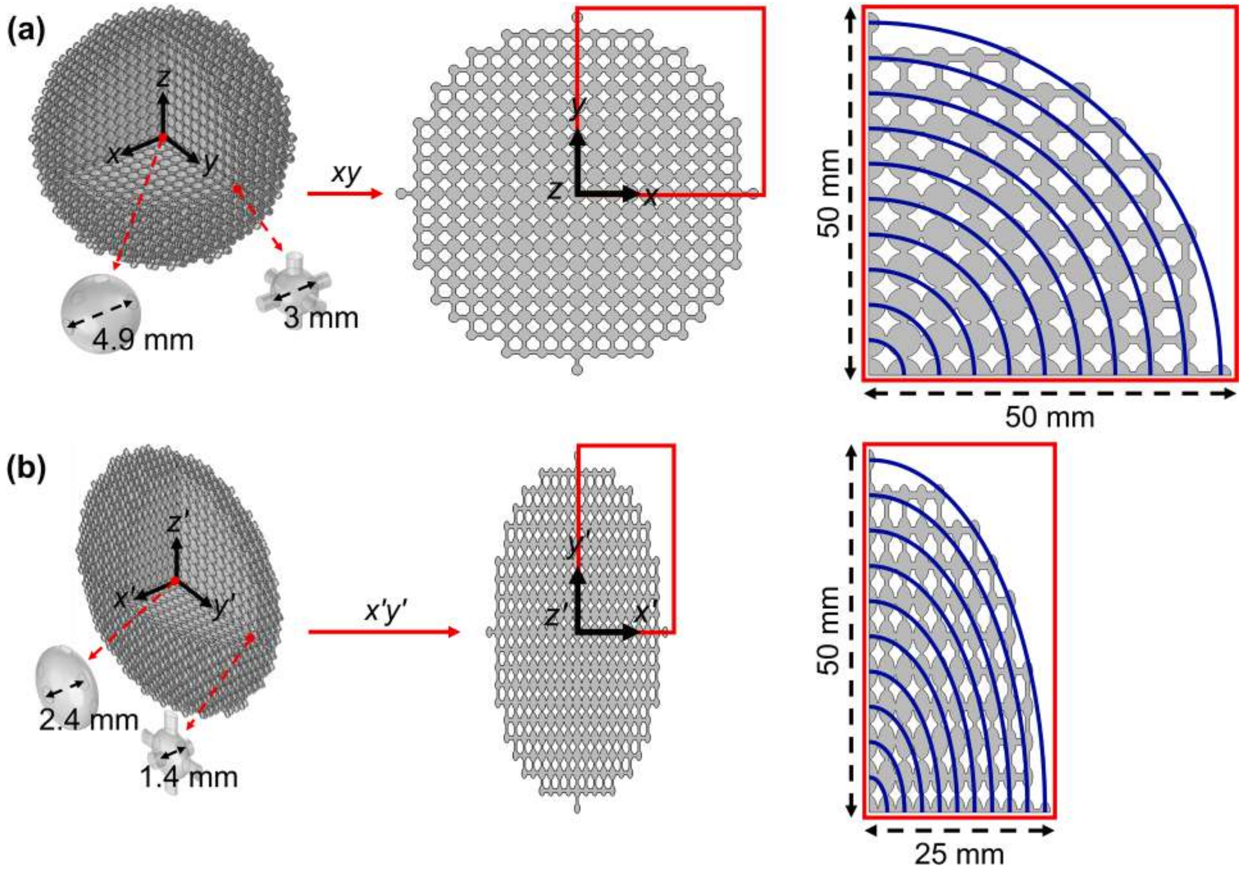

Fig. 7. The designed (a) spherical and (b) ellipsoidal lenses with 10 layers, and the meta-atoms (largest and smallest).

Table 1
Dimension of the spherical and ellipsoidal lenses.

Layer number	Radius of layer (mm)		Diameter of meta-atom (mm)	
	Spherical lens	Ellipsoidal lens	Spherical lens (D)	Ellipsoidal lens ($D_x'/D_y'/D_z'$)
0	0	0	4.9	2.4/4.9/4.9
1	5	2.5	4.9	2.4/4.9/4.9
2	10	5	4.9	2.4/4.9/4.9
3	15	7.5	4.8	2.4/4.8/4.8
4	20	10	4.8	2.3/4.8/4.8
5	25	12.5	4.7	2.3/4.7/4.7
6	30	15	4.6	2.2/4.6/4.6
7	35	17.5	4.4	2.1/4.4/4.4
8	40	20	4.2	1.9/4.2/4.2
9	45	22.5	3.8	1.7/3.8/3.8
10	50	25	3.0	1.4/3.0/3.0

$$\begin{cases} B_{eff} = \dfrac{Z_{eff}}{n_{eff}} \\ \rho_{eff} = Z_{eff} \times n_{eff} \end{cases}. \tag{9}$$

The effective acoustic material properties (speed of sound, acoustic impedance, bulk modulus, and mass density) are shown in Fig. 6.

Fig. 6 shows the calculated effective acoustic parameters using the S-parameter retrieval method. The largest inclusions were selected for the retrieved values because the larger the volume of meta-atoms, the lower the PBG frequency, as shown in Figs. 3(b) and (d). The imaginary parts of the effective material properties are because of the diffuse scattering loss of the scatterers. In Fig. 6(a), the real parts of the effective speed of sound c_{eff} change from positive to negative at frequencies 100 kHz and 160 kHz, respectively. That is,

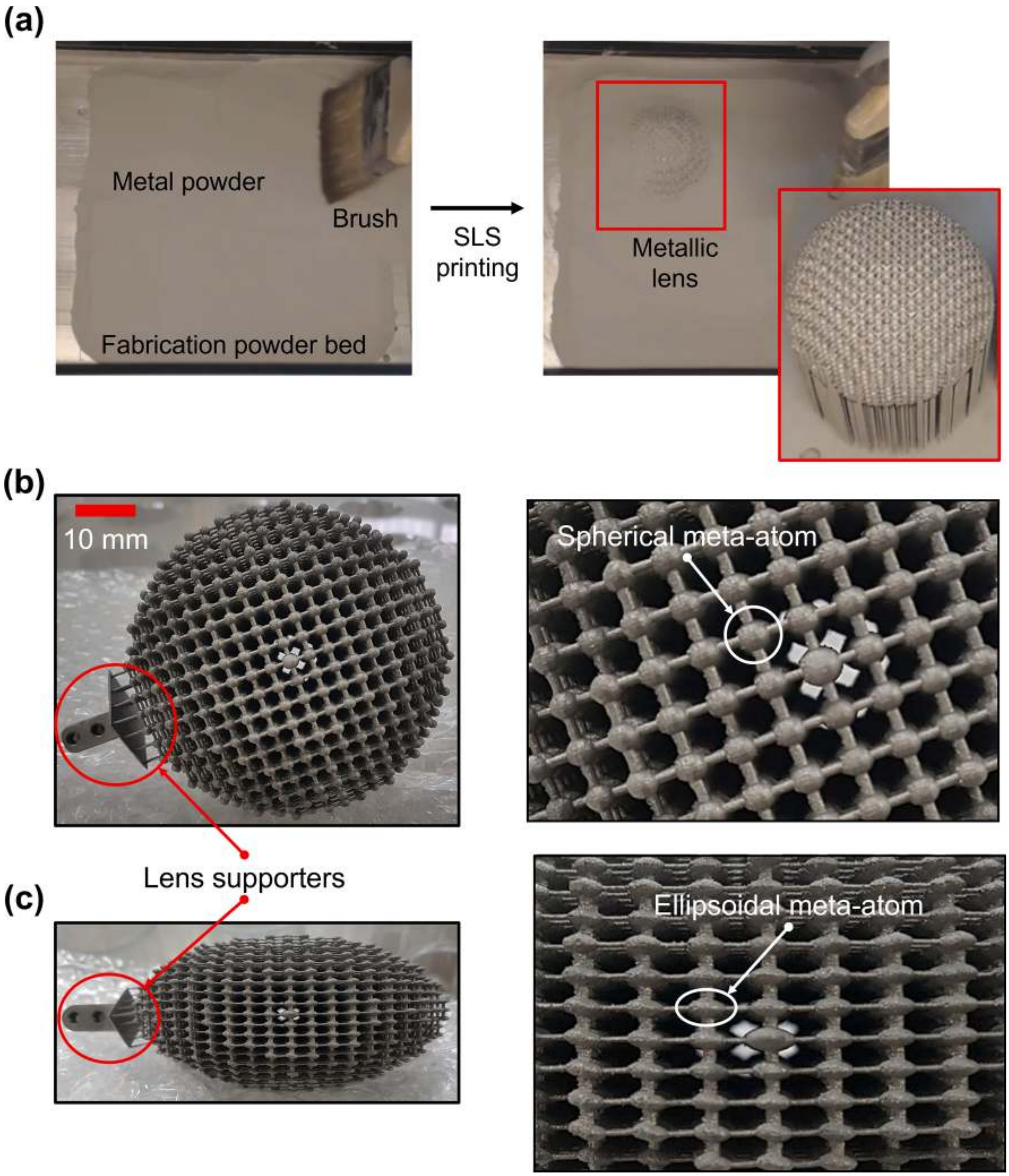

Fig. 8. (a) Picture of the final fabrication process for the spherical lens. The fabricated (b) spherical and (c) ellipsoidal lenses combined with the lens supporters that can be fixed to the testbed.

ultrasound undergoes negative refraction at these frequencies. In the available frequency, the higher the frequency, the greater the reflection value (S_{11}) than the transmission value (S_{21}), as shown in Figs. 5(c) and (d). Until the maximum PBG frequency, 100 kHz and 160 kHz, Z_{eff}, B_{eff}, and ρ_{eff} in Figs. 6(b)–(d) are increased while the frequency increases. However, these values are near zero in the PBG frequency range. The spherical and ellipsoidal meta-atoms have negative effective parameters because the acoustic waves are completely blocked near 100 kHz and 160 kHz, respectively. These parameters are designed with nonresonant scatterers, and the effective parameters calculated by the S-parameter retrieval method agreed with the results of the PC band structure calculations analysis in Figs. 3(c) and 5(b). For the designed acoustic metamaterials, the regions corresponding to PBG, and negative properties are excluded from the available frequencies.

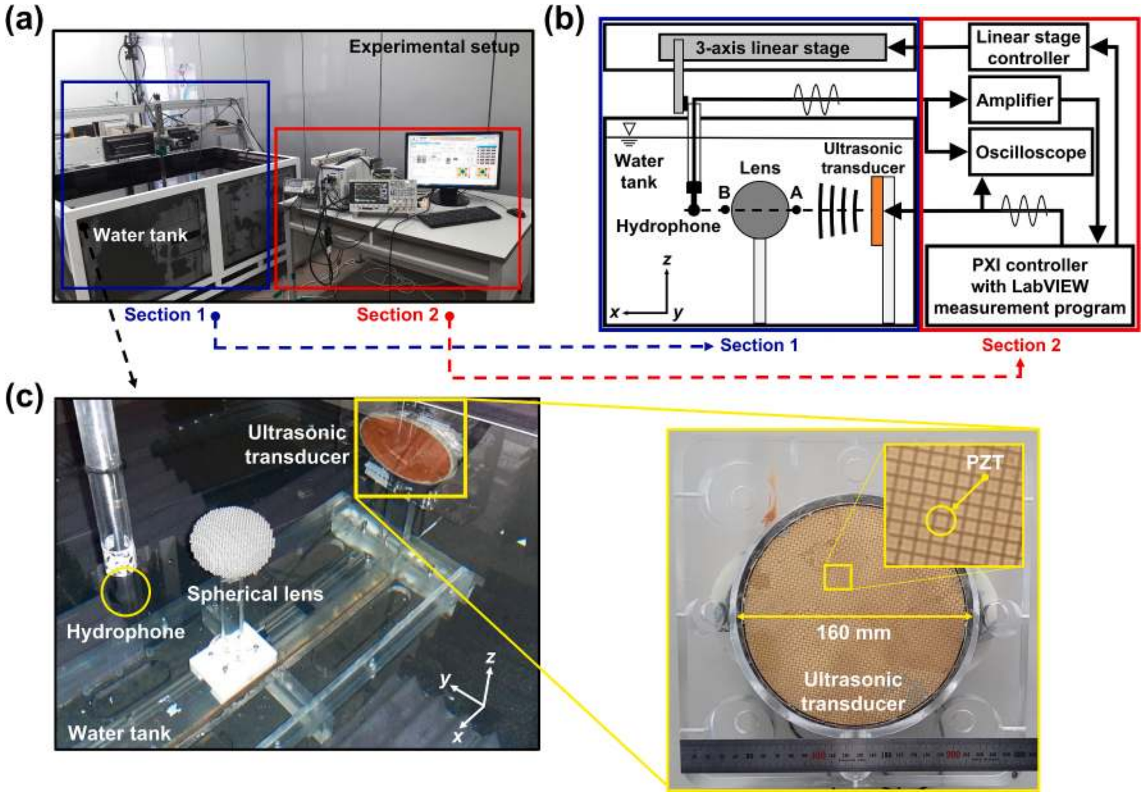

Fig. 9. (a) Experimental setup and (b) its schematic diagram. Several pieces of equipment such as a three-axis linear stage and its controller, PXI controller with LabVIEW measurement program, oscilloscope, hydrophone, transducer, and amplifier are used. (c) Setting for measuring underwater ultrasound using hydrophone, spherical lens, and ultrasonic transducer in the water tank.

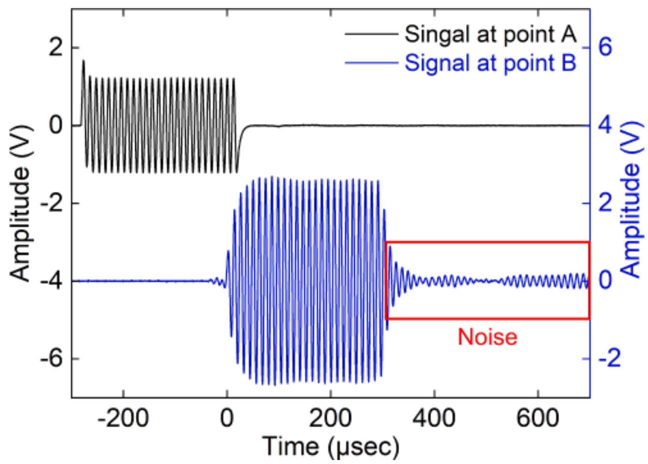

Fig. 10. Incoming burst signal at the point A and outgoing burst signal at the point B. The frequency is 80 kHz.

4. Underwater ultrasound imaging

4.1. Design, fabrication, and experimental setup

Impedance matching is required to reduce energy loss. The approach to achieve impedance matching is to increase the number of layers, that is, the more number of layers, the lower would be the energy loss. However, an appropriate number of layers should be considered to reduce computational cost and facilitate fabrication. Thus, the designed spherical and ellipsoidal lenses are discretized

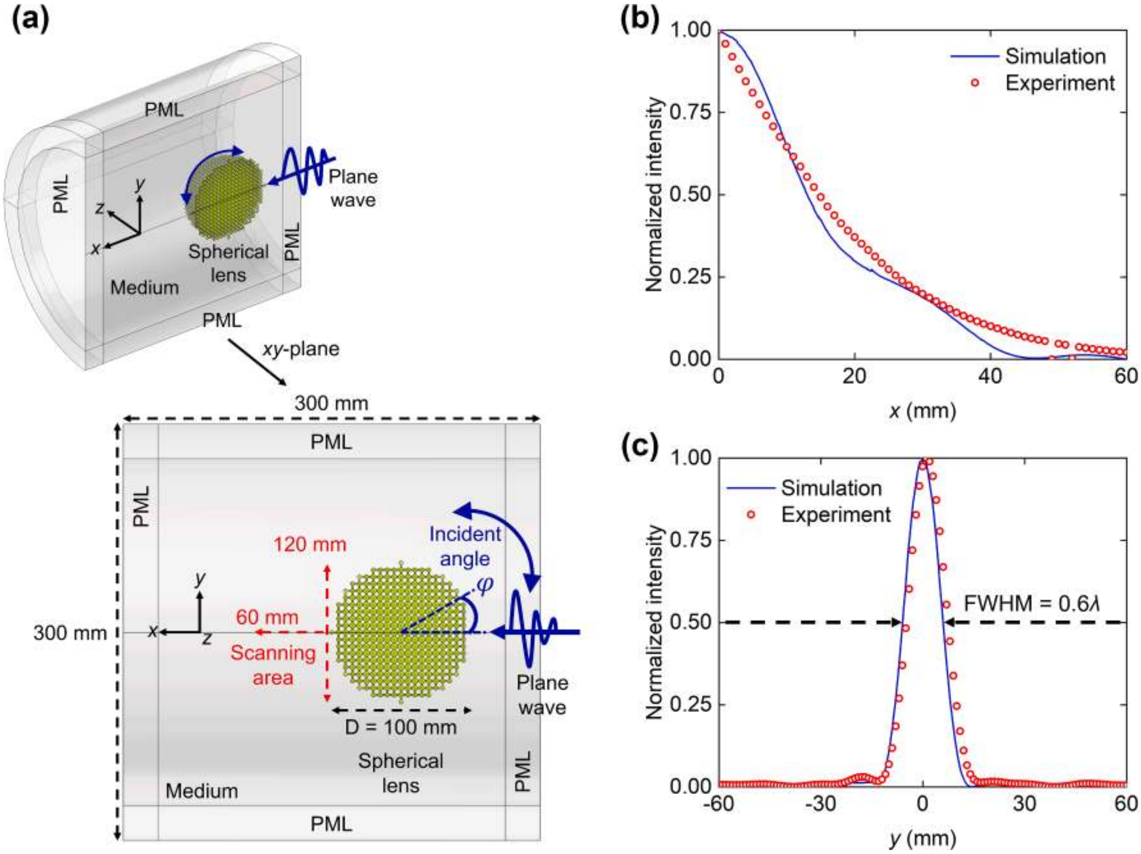

Fig. 11. (a) Simulation modeling. Normalized intensity at 80 kHz along the (b) x- and (c) y-axes, respectively.

and divided into 10 concentric layers, as shown in Figs. 7(a) and (b), respectively. The dimension of the lenses is shown in Table 1. The refractive indices by layers are decreased from the lens' center to the surface. The mass densities of the unit cell of the lenses have been decreased by raidal distance in the x-, y-, and z-axes directions. Picture of the final fabrication process for the spherical lens using a metal 3D printer, Concept laser M2 cursing, made by GE Additive [57], is provided in Fig. 8(a), and the fabricated spherical and ellipsoidal lenses made of stainless steel are shown in Figs. 8(b) and (c), respectively. The lens supporters (area circled in red line) are added at the bottom of the lenses to fix the lenses to the testbed frame in the water tank. The fabrication resolution is 0.1 mm.

The experimental setup and schematic diagram are shown in Figs. 9(a) and (b), respectively. The hydrophone (Bruel & Kjaer type 8105), amplifier (Bruel & Kjaer type 2693), ultrasonic transducer, and three-axis linear stage (MISUMI XCVL6300) with a linear stage controller (SURUGA SEIKI D520) are controlled by LabVIEW measurement program of PXI controller (NI PXIe-1062Q chassis with PXIe-6124 multifunction DAQ and PXIe-5413 AWG). When the ultrasonic wave is generated and propagated from the transducer to the lens in the water tank, the acoustic pressure data acquired by the hydrophone was recorded by the LabVIEW measurement program for the postprocessing of the data.

The experiments were conducted in the water tank surrounded by sound-absorbing materials to reduce the reflection of ultrasonic waves, as shown in Fig. 9(c). The distance between the ultrasonic transducer and the lens is 200 mm ($\gg 10\lambda$), and the diameter of the transducer is 160 mm which is larger than that of the lenses to generate plane waves. The center of the lenses was aligned to the center of the ultrasonic transducer. The ultrasonic transducer is not focalized, but its top surface is flat, as shown in Fig. 9(c). It is made of 1–3 composite type lead zirconate titanate (PZT) by Ceracomp. Co. in Republic of Korea [58].

The burst signals at the points A and B in Fig. 9(b) are shown in Fig. 10. The pulse duration was set to minimize the noise of the outgoing burst signal, which was reflected by the wall, as shown in Fig. 10. The signal at the point B is enhanced by two times larger than that at the point A by the focusing effect of the lens.

4.2. Numerical and experimental results

The numerical simulation was conducted by COMSOL Multiphysics V5.3, as shown in Fig. 11(a). The structural boundary condition was that the meta-atom located at the center of the lens is fixed in the x-, y-, and z-axes. The plane wave was generated from the right wall and propagated to the left wall via the lens. PMLs were applied around the lens. By rotating the lens around the z-axis, the focus imaging performance of the lens by an incident angle was tested. The normalized intensity and FWHM were scanned along the y- and x-

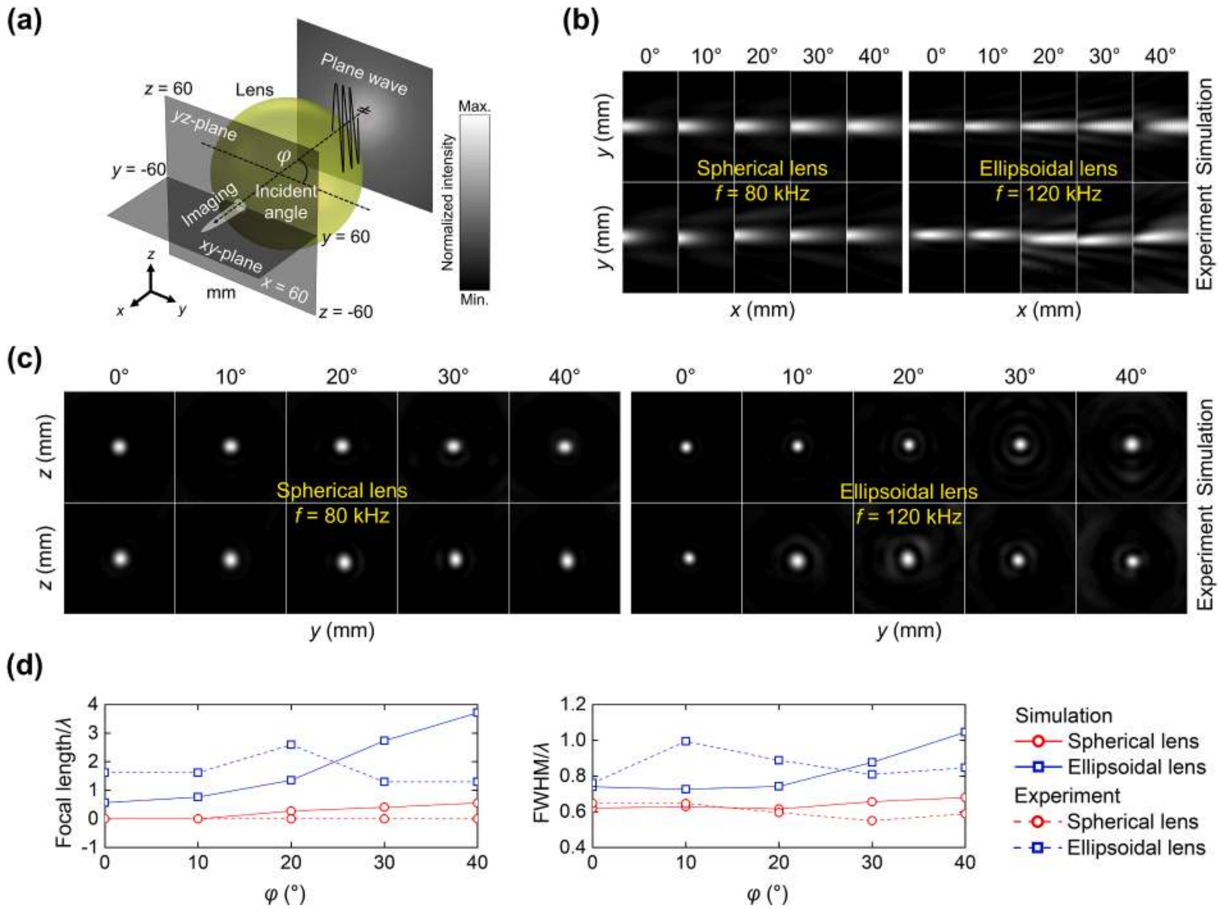

Fig. 12. (a) Schematic diagram for the experiments underwater. Underwater ultrasound imaging of numerical and experimental results of the spherical lens (f = 80 kHz) and the ellipsoidal lens (f = 120 kHz) in the (b) *xy*- and (c) *yz*-planes, respectively. (d) Numerical and experimental results of the focal length and FWHM by an incident angle φ.

axes directions, respectively, as shown in Figs. 11(b) and (c). The numerical and experimental results were in good agreement. Herein, it is introduced representatively for the spherical lens.

The underwater acoustic lenses were simulated as shown in Fig. 11, fabricated in Figs. 7 and 8, and experimented on Figs. 9 and 10 in water ($\rho = 998$ kg/m^3 and $c = 1{,}481$ m/s). The scanned area is shown in Figs. 11(a) and 12(a). The lenses were analyzed to obtain the focal length and the FWHM in the scanning area. The focal length is defined as the distance from the surface of the lenses to the point where the highest sound intensity is obtained along the center line of the lenses in the *x*-axis direction. The FWHM is defined as the width of the spectral sound intensity curve obtained between the points on the *y*-axis of the sound intensity amplitude, which is half of the maximum sound intensity, $I_{max}/2$. All values of the sound intensity are scaled between 0 and 1, and the white region indicates the area where the ultrasound is focused. When the plane waves are normally incident on the lenses, the *xy*- (60 × 120 mm^2) and *yz*- (120 × 120 mm^2) planes were scanned to obtain the sound pressure and the images. The beam steering performance of the lens was tested by rotating the lens instead of changing the orientation of the transducer with the incident angles φ of the lenses from 0° to 40° in 10° steps.

Figs. 12(b) and (c) show the imaging of sound intensity in the *xy*- and *yz*-planes of the spherical (f = 80 kHz) and ellipsoidal (f = 120 kHz) lenses by incident angle φ at the focal point where it is observed the maximum sound intensity. The values in Figs. 12(b) and (c) are expressed as normalized intensities, which is increased from black as minimum value to white as maximum value. The normalized sound intensity of the ellipsoidal lens is not focused on the surface of the lens, but there is a deviation from the center. However, the spherical lens shows uniform performance in all directions (omnidirectionality). Figs. 12(b) and (c) (especially in the *xy*-plane) show that the spherical lens has a better beam steering performance than that of the ellipsoidal lens. The graph demonstrating these results in more detail is shown in Fig. 12(d). The ellipsoidal lens has a longer focal length than that of the spherical lens (0λ to 0.5λ). The FWHM of the spherical lens is almost flat near 0.6λ.

In Fig. 13, the frequency-dependent imaging results of the ellipsoidal lens were analyzed experimentally. Figs. 13(a) and (b) show imaging in the *xy*- and *yz*-planes at frequencies between 60 kHz and 160 kHz in 20 kHz steps, respectively, when the incident angle φ is zero. As indicated by the previously analyzed effective acoustic parameters, the ultrasound imaging was achieved at the available

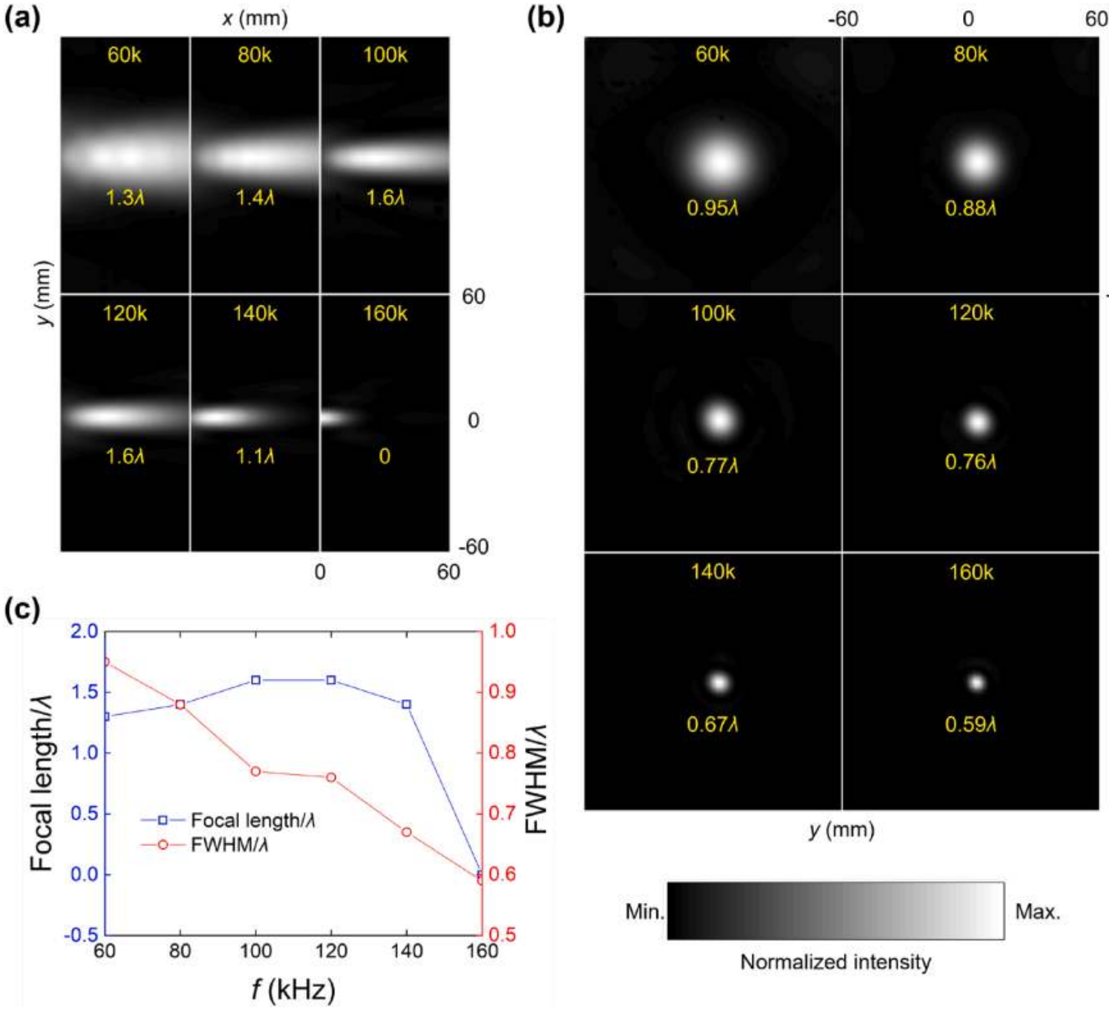

Fig. 13. Experimental results of the underwater ultrasound imaging of the ellipsoidal lens in the (a) *xy*- and (b) *yz*-planes, respectively. Measurements were performed at frequencies between 60 kHz and 160 kHz in 20 kHz step for an incident angle φ of zero. (c) Experimental results for the focal length and FWHM by frequency. The blue line-hollow square and red line-hollow circle represent the focal length and FWHM, respectively.

frequency range from 60 kHz to 160 kHz. The focal length and FWHM of the ellipsoidal lens were 1.5λ to 0λ and 0.95λ to 0.59λ at frequencies between 60 kHz and 160 kHz, respectively. The values of the FWHM is decreased while the frequency is increased, as shown in Fig. 13(c). Waves are refracted more at high frequencies than that at low frequencies. As the frequency increases, the refractive index of the designed lens approaches more closely to that of the Luneburg and TO-based lens. Thus, higher resolution imaging can be obtained at high frequency. Consequently, the TO-based lens has a wide bandwidth at high frequencies for underwater ultrasonic waves.

5. Conclusion

In this study, the numerical simulations and experiments of the 3D acoustic metamaterial lenses based on the Luneburg lens and transformation optics were performed underwater. Two shapes of lenses were considered: spherical and ellipsoidal. The dispersion curves, equi-frequency contours, and effective material properties were determined using the phononic crystal band structure calculations and scattering parameter retrieval method, and the Bloch modes were in good agreement. The equi-frequency contours confirmed that the wavevectors underwent positive refraction, and frequency ranges were identified in detail. Finally, the ultrasound imaging was achieved numerically and experimentally in underwater condition.

Further study is required on the following aspects:

1) Stainless steel is used to achieve a high impedance difference between the lenses and water. However, air inclusions may be an alternative. Air enters the inside of scatterers, which is ideal as it can generate a high refractive index even if the structure is not metallic [32,33]. It is also an economical choice in terms of cost.

2) Commercializing these lenses immediately would be challenging; the lenses are difficult to fabricate because of the complex structure. This is a limitation of gradient index lenses (and probably all metamaterials). Thus, it is necessary to design a structure that is as simple as possible while still maintaining effective performance.
3) It is difficult to fabricate the lenses as designed by fabrication resolution, and aberrations inevitably occur in experimental results. However, this problem could be solved owing to the development of manufacturing technology over time.
4) The beam steering performance of the ellipsoidal lens is deteriorated, as shown by the EFCs (or EFSs) in Fig. 4(b) and the imaging in Fig. 12. The imaging performance may be limited in the case of a high incident angle such as 90°. This is inevitable because it deviates from the spherical shape, which has the same performance for all incident angles.
5) The speed of sound is a function of temperature. However, in this study, the lenses were simulated and experimented at room temperature. Therefore, future studies on the performance by water temperature variation may be necessary.

However, the advantages of each lens are clear and should be emphasized. The spherical lens is relatively bulky compared to the ellipsoidal lens but is omnidirectional and does not suffer from aberration. The ellipsoidal lens would be suitable for broadband devices because the available frequency range is broader than that of the spherical lens. If various structures suitable for fabrication and use are proposed by applying transformation optics lenses, these lenses are expected to have prospects in imaging applications requiring a wide field of view such as medical imaging systems, nondestructive evaluation, and sound navigation ranging in the future.

CRediT authorship contribution statement

Jung-Woo Kim: Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Gunn Hwang:** Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – review & editing. **Seong-Jin Lee:** Data curation, Formal analysis, Software, Investigation, Writing – review & editing. **Sang-Hoon Kim:** Funding acquisition, Conceptualization, Methodology, Supervision, Validation, Writing – review & editing. **Semyung Wang:** Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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